

Data Flow Diagram

Date	28 JUNE 2026
Team ID	xxxxxx
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	2 Marks

Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	Farmer – Provides soil and environmental data and receives crop recommendations.
Rectangle with numbered header	Process	• Data Input & Validation • Data Preprocessing • Machine Learning Prediction Engine • Display Recommendation (Flask Web Application)
Rectangle (solid fill, no number)	Data Store	• Soil & Crop Dataset • Prediction Results Database
Labeled arrow	Data Flow	• Soil Parameters (N, P, K, Temperature, Humidity, pH, Rainfall) • Validated Input • Processed Data • Crop Prediction • Recommendation Result • Crop Suitability Information

Example

The diagram below illustrates how the symbols above come together to form a complete data flow diagram



